Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Abstract

We consider the problem of covering/connecting a multi-interface network with a minimum cost assignment of interfaces to vertices. In the covering problem we demand that every edge is covered by at least one common interface assigned to its endpoints. In the connection problem we are given a set of terminals $U \subseteq V$ and we demand that the subgraph of activated edges spans all of them. We give an $O(\log m)$ approximation algorithm for the min-max cost criterion. The algorithm is based on rounding the optimal solution of a natural, set-cover-style linear programming relaxation of the problem. We also show how to use this LP to obtain a k-approximation algorithm, where k is the number of interfaces. Importantly, both of the aforementioned procedures work even when the cost of activating an interface depends on the interface as well as the vertex to which it is assigned. This improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation, where b is a certain parameter of the input instance which can be as large as $\Omega(n)$ and $c_{\max}$ is the maximum cost of activating an interface and $O(k)$-approximation for the uniform cost case [Algorithmica 2012]. For the connection problem we devise an $O\big(\log^2 m\big)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem. The algorithm is based on a similar LP relaxation with additional cut constraints to ensure connectivity of the terminals.

1 Introduction

Designing Internet of Things (IoT) networks often demands establishing connection between large number of devices, which can be achieved by choosing between large variety of interfaces available at each device. Moreover, the devices present in the network may vary according to the interfaces they provide for the usage. For example some device may be able to connect by Bluetooth and Wi-Fi, while the others may provide Zigbee and Z-Wave interfaces. Since IoT devices are typically resource-constrained, it may be desirable to try to establish every relevant connection, while minimizing the amount of resources required to do so.

We model the above problem as a graph problem. Every device in the network is modeled by a vertex with a set of labels representing its available interfaces. Two devices can communicate if they share a common available interface and are sufficiently close to each other, and the edges of the graph represent the possible connections. A connection is established between two devices when they share a common active interface. We want to find an assignment of active interfaces so that we ensure either coverage or connectivity of the network. In the coverage problem, we wish to establish every possible connection in the network (see Figure 1 for an example) while in the connectivity problem, we are additionally given a subset of vertices called terminals, an we require that terminals belong to the same connected component in the subgraph induced by active interfaces (see Figure 2 for an example). In both problems, we assume that activating an interface induces a cost that depends on the type of interface and on the vertex that activates it. The goal is to minimize either the min-sum cost (the sum of costs over all vertices) or the min-max cost (the maximum cost in a vertex).

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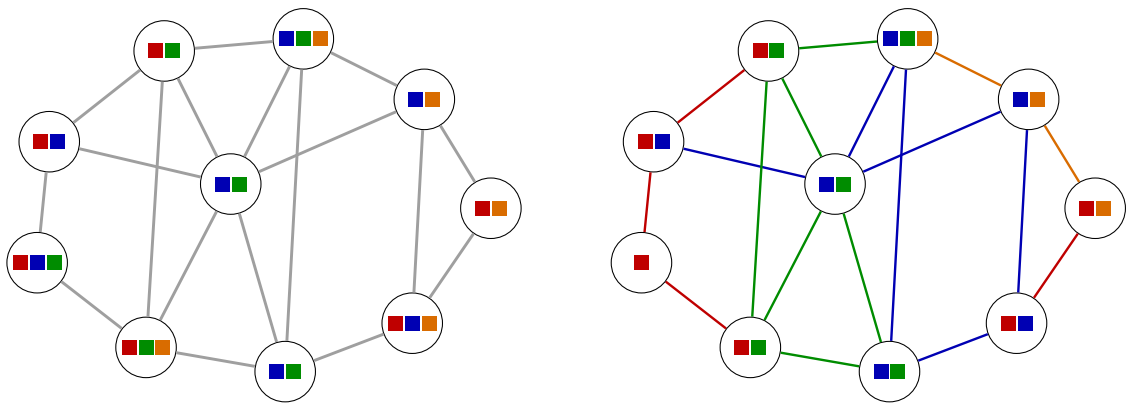
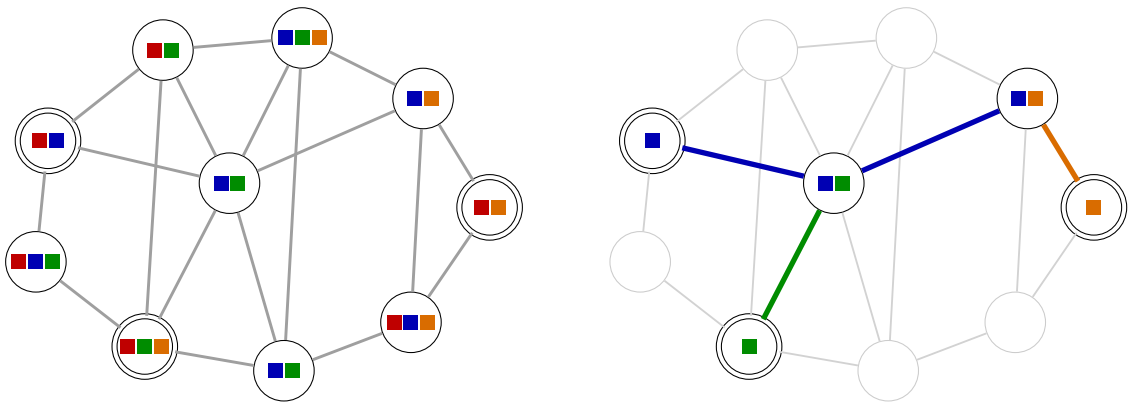


Available labels

Example covering assignment

Figure 1: An example instance of a multi-interface network with available interfaces at each vertex and a feasible covering assignment.



Available labels (double contour = terminal)

Example connecting assignment

Figure 2: An example instance of a multi-interface network with available interfaces at each vertex and a feasible connecting assignment.

1.1 Related Work

In the following, we denote k the number of different interfaces across the network, n the number of vertices, Δ the maximum degree of the graph, $c_{\max}$ the maximum cost and $c_{\min}$ the minimum cost of an interface. In the *homogeneous* setting, the costs depend on the interface only while in the *heterogeneous* case they depend on the interface and on the vertex.

Coverage Problem. The algorithmic study of problems in multi-interface networks started with the *coverage* problem, introduced in Caporuscio et al. [6]. The coverage problem has been studied only in the homogeneous setting, both for min-sum [13] and min-max [8] objective functions.

The min-sum variant is APX-hard for any constant $k \geq 3$ and unit costs, and it is hard to approximate within a $O(\ln k)$ factor even with unit costs, but it has a $\min\{k-1, \sqrt{n}(1+\ln n)\}$-approximation [13].

The min-max variant is NP-Hard for any constant $\Delta \geq 5$, $k \geq 16$ and for unit costs, and it is not approximable within a $\eta \ln \Delta$ factor for some constant η even on trees and with unit costs. On the positive side, it admits a $(\ln \Delta + 1 + b \cdot \min\{c_{\max}, \ln \Delta + 1\})$-approximation, where b is a parameter that depends on the structural properties of the input graph and is bounded by many other graph structutal parameters such as treewidth, arboricity or maximum degree. It also has a $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$-approximation and a $(\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}})$-approximation [8].

Connectivity Problem. The *connectivity* problem was introduced by Kosowski et al. [15]. It was first studied in the homogeneous setting and in the special case where all vertices are terminals, both for the min-sum [15] and the min-max objective functions [8].

The min-sum variant with terminals only and homogeneous costs is APX-hard for $k \geq 2$, $\Delta \geq 4$ and unit costs already, but has a 2-approximation [15]. This result was improved by Athanassopoulos et al. [4] to a randomized $(\frac{3}{2} + \epsilon)$-approximation for any constant ϵ. Moreover, this approximation algorithm applies to the more general problem where for each interface, we are additionally given the set of edges that it can cover.

The min-max variant where costs are homogeneous and all vertices are terminals is NP-Hard for any constant $\Delta \geq 3$, $k \geq 10$ and for unit costs. Similarly to the coverage problem, it is not approximable within a $\eta \ln n$ factor even for unit costs, but has simple $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$- and $\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}}$-approximations [8].

Athanassopoulos et al. [4] expand the study of the min-sum connectivity problem to *group connectivity*, a more general problem where the solution graph should connect together groups of vertices that might not span the whole set of vertices. They provide a randomized 4-approximation for it. They also initiate the study of this problem in the heterogeneous setting, where costs can also depend on the vertex. They show that the heterogeneous connectivity problem is hard to approximate within $o(\ln n)$, but the group connectivity heterogeneous case admits a $O(\ln n)$-approximation.

Other Multi-interface Problems. There has been some work on the study of the coverage problem from the perspective of parameterized complexity, possibly with an addtional constraint on the maximum number of active interfaces in a vertex [2, 1, 5], yielding efficient algorithms in some graph classes. Some papers also study other problems in multi-interface networks, like cheapest path [15], matchings [14], or flow problems [9].

1.2 Our contribution and techniques

In this paper we give a first $O(\log m)$-approximation algorithm for the min-max connection problem with non-uniform costs. This improves the previously known result of [8], which gave an $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation for the min-max covering version, where b is the parameter related to the existence of a b-owner function [10, 7]. This als improves the $O(\sqrt{n} \cdot \log n)$-approximation for the min-sum covering problem by [13]. Our approach is based on a set-cover-style LP relaxation together with a randomized rounding. We also show how to use the same LP relaxation to obtain a k-approximation algorithm for the min-sum and min-max covering problems, where k is the number of interfaces. All these guarantees hold in the non-uniform setting where activation costs depend on both the interface and the vertex it is assigned to.

2 Preliminaries

Let $G = (V, E)$ be a connected simple graph. Let $[k]$ be the set of different types of interfaces across the network and let $\lambda\colon V \to [k]$ be a function that assigns to each vertex $v \in V$ its set of available interfaces such that for each edge $uv \in E$, $\lambda(u) \cap \lambda(v) \neq \emptyset$. An assignment of active interfaces is a function $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$. We say that an edge $uv \in E$ is *covered* if $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$ and we denote by G_A the subgraph of G induced by covered edges.

In the *coverage problem*, we wish to find an assignment of active interfaces λ_A such that $G_A = G$. We call such an assignment a *covering assignment*.

In the *connectivity problem*, we are additionally given $p \geq 1$ pairwise disjoint groups of *terminals* $U_1, U_2, \ldots, U_p \subseteq V$ and we wish to find an assignment of active interfaces λ_A such

that for every $r \in [p]$, all vertices in U_r lie in the same connected component of G_A. Vertices in $V \setminus \bigcup_{r=1}^{p} U_r$ may be used as Steiner vertices. We call such an assignment a *connecting assignment*. The special case $p = 1$ with $U_1 = V$ corresponds to requiring that G_A is a connected spanning subgraph of G.

We are also given a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, which denotes the cost of activating the interface $i \in [k]$ in the vertex $v \in V$. The sum cost of an assignment λ_A is defined as:

$$\texttt{COST}_S = \sum_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)$$

and the max cost of and assignment λ_A is defined as:

$$\texttt{COST}_M = \max_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

The goal of the min-sum (resp. min-max) assignment problem is to find an assignment λ_A that minimizes $\texttt{COST}_S$ (resp. $\texttt{COST}_M$). The considered problems are formalized below:

Coverage Problem

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to [k]$, a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$.
Task: Find a min-sum (resp. min-max) covering assignment λ_A minimizing $\texttt{COST}_S$ (resp. $\texttt{COST}_M$).

Connectivity Problem

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to [k]$, a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, pairwise disjoint groups of terminals $U_1, \ldots, U_p \subseteq V$.
Task: Find a min-sum (resp. min-max) connecting assignment λ_A minimizing $\texttt{COST}_S$ (resp. $\texttt{COST}_M$).

Additionally, we will need the following probability facts:

Theorem 2.1 (Markov's Inequality)**.** *Let X be a non-negative random variable and $a > 0$. Then:*

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 2.2 (Wald's Identity)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite expectation $\mathbb{E}[X]$, and let N be a non-negative integer-valued random variable independent of the sequence $(X_i)_{i \geq 1}$ with finite expectation $\mathbb{E}[N]$. Then:*

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[N] \cdot \mathbb{E}[X].$$

Theorem 2.3 ((weighted) Chernoff Bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables taking values in $[0, 1]$, let $w_1, w_2, \ldots, w_n$ be non-negative weights, such that for every $i \in [n]$ we have that $w_i \in [0, 1]$. Let $X = \sum_{i=1}^{n} w_i \cdot X_i$ and let $\mu = \mathbb{E}[X]$. Then for every $\delta_1 > 0$ and $\delta_2 \in (0, 1)$ we have that:*

$$\Pr(X \geq (1 + \delta_1) \cdot \mu) \leq \left(\frac{e^{\delta_1}}{(1 + \delta_1)^{1+\delta_1}}\right)^{\mu} \qquad \Pr(X \leq (1 - \delta_2) \cdot \mu) \leq \left(\frac{e^{-\delta_2}}{(1 - \delta_2)^{1-\delta_2}}\right)^{\mu}.$$

The proof of the above fact follows the proof of the standard Chernoff bound and we include it in the appendix for completeness.

3 $O(\log n)$-approximation algorithms for the covering problem

Below we show an $O(\log m)$ approximation algorithms for all four variants and $O(k)$-approximation for the covering problems. The idea of our ILP formulation and the algorithm is inspired by the rounding procedure known for set cover [16] and related problems [3]

3.1 Rounding the costs

Due to technical reasons, we show how to produce a polynomial number of instances, with cost function bounded by a polynomial in n. By testing consecutive powers of 2, we find the smallest number b, such that the costliest activated interface in the optimal solution has cost at most 2^b. For each vertex $v \in V$, we discard all of the interfaces from $\lambda(v)$ with cost larger than 2^b and we normalize the costs of remaining interfaces by dividing them by 2^b. Note that the number of guesses is polynomial. We also can assume that $\texttt{OPT} \geq 1/2$, since the cost of any solution is at least the cost of activating one interface, which is at least 1/2 by our rounding.

3.2 ILP formulation of covering multi-interface networks

We start with the following ILP formulation of the min-sum covering assignment problem. Let $x_{i,v}$ be a binary variable that is 1 if and only if the interface $i \in \lambda(v)$ is assigned to vertex $v \in V$. We consider two types of vertices. For a given vertex $v \in V$, if $\sum_{i\in\lambda(v)} c(i,v) \leq 1$, then we can safely assume that all interfaces in $\lambda(v)$ are activated at v, since the cost of activating all of them is at most 1 and thus at most $2 \cdot \texttt{OPT}$. We call such vertices *cheap*. For the remaining vertices, we call them *expensive*. Note that for every expensive vertex v, we can assume that $\sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$, since this will always result in a solution of cost at most $2 \cdot \texttt{OPT}$. Thus, we can write the following ILP formulation of the min-max covering assignment problem:

$$
\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i\in\lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i\in\lambda(v)} c(i,v) \cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v)
\end{aligned}
$$

Note that the above formulation is not linear due to the presence of the min function. However, by adding an additional variable $y_{i,uv}$ we can linearize the constraints of type $\sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1$ by replacing them with $\sum_{i\in\lambda(u)\cap\lambda(v)} y_{i,uv} \geq 1$ together with the constraints $y_{i,uv} \leq x_{i,u}$ and $y_{i,uv} \leq x_{i,v}$ for every $i \in \lambda(u) \cap \lambda(v)$ and $uv \in E$. Thus, we can solve the LP relaxation of this formulations in polynomial time.

3.3 The algorithm

Algorithm 1: A relaxation-based $O(\log m)$-approximation algorithm.

1 Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v),v\in V}$.
2 Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
3 $\lambda_A(v) \leftarrow \{i \in \lambda(v) \colon 3 \ln m \cdot x_{i,v} \geq t_i\}$, for every $v \in V$.
4 **return** $\{\lambda_A(v)\}_{v\in V}$.

Theorem 3.1. *The above algorithm is an $O(\log m)$-approximation algorithm for the min-max covering assignment problems which succeeds with high probability.*

Proof. We have the following lemmas:

Lemma 3.2. *The algorithm returns a solution of cost at most $9 \ln m \cdot \texttt{OPT}$ with high probability.*

Proof. Fix a vertex $v \in V$. We want to show that with high probability, the cost of activating interfaces at v is at most $9 \ln m$ times the cost of activating interfaces at v in the fractional solution. If v is cheap, then the cost of activating all interfaces at v is at most 1, so the claim holds. Assume that v is expensive. By $X_{v,i}$ denote the random variable that is 1 if $3 \ln m \cdot x_{i,v} \geq t_i$ and 0 otherwise. Then we can write $c(\lambda_A(v)) = \sum_{i \in \lambda(v)} X_{v,i} \cdot c(i,v)$. We have: $\mathbb{E}[X_{v,i}] = \Pr[i \in \lambda_A(v)] = 3 \ln m \cdot x_{i,v}$. Thus, $\mathbb{E}[c(\lambda_A(v))] = 3 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v)$. By applying the Chernoff bound with $\delta = 2$ we get that

$$\Pr[c(\lambda_A(v)) \geq 3 \cdot \mathbb{E}[c(\lambda_A(v))]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[c(\lambda_A(v))]} \leq \exp\{-\mathbb{E}[c(\lambda_A(v))]\} \leq \frac{1}{m^3}$$

where the last inequality is by using the fact that, $\sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$. By applying the union bound over all n vertices we get that with probability at least $1 - \frac{n}{m^3} \geq 1 - \frac{1}{m}$ (assuming $m \geq 2$) for every $v \in V$, $c(\lambda_A(v)) \leq 9 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v)$. Thus, with high probability we have $\texttt{COST}(\lambda_A) \leq 9 \ln m \cdot \max_{v \in V} \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) = 9 \ln m \cdot \texttt{OPT}$. □

Lemma 3.3. *With high probability, the above algorithm returns a covering assignment.*

Proof. Consider an edge $uv \in E$. We have:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-3 \ln m \cdot \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\{-3 \ln m\} = \frac{1}{m^3}
\end{aligned}$$

By applying the union bound over all m edges we get that the algorithm returns a covering assignment with probability at least $1 - \frac{m}{m^3} = 1 - \frac{1}{m^2}$. □

Combining the above lemmas gives the claim. □

4 $O(\log^2 n)$-approximation algorithm for the connection problem

4.1 ILP formulation

In this section we again assume that $\texttt{OPT} \geq 1/2$, the cost of activating any interface is at most 1 and we distinguish between cheap and expensive vertices as before. We use a similar LP to model the connection problem, however we use additional flow variables to encode the connectivity requirement. The $x_{i,v}$ variables are defined as before. For each edge e we add an additional flow variable y_e which is 1 if and only if the edge e is covered by the solution. Connectivity within each group U_r is enforced via cut constraints: for every $r \in [p]$ and every cut $(S, V \setminus S)$ that

separates at least one pair of terminals in U_r, the total y-weight of edges crossing the cut must be at least 1. Then we can formulate the min-max connection problem as follows:

$$
\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& y(\delta(S)) \geq 1 \quad \forall r \in [p],\ \forall S \subset V \colon S \cap U_r \neq \emptyset,\ (V \setminus S) \cap U_r \neq \emptyset \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E
\end{aligned}
$$

Observe that the above ILP has exponential number of constraints, but we can solve its LP relaxation in polynomial time by using the ellipsoid method and a separation oracle for the cut constraints, which can be implemented using a max-flow algorithm.

4.2 The algorithm

Algorithm 2: A relaxation-based $O(\log^2 m)$-approximation algorithm.

Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}, \{y_{uv}\}_{uv \in E}$.
For every edge $e \in E$, let $p_e = \min\{1, 4 \ln m \cdot y_e\}$.
With probability p_e set $Y_e = 1$, otherwise set $Y_e = 0$.
Let H be the subgraph of G induced by the edges e such that $Y_e = 1$.
$\mathcal{S} \leftarrow \emptyset$.
$j \leftarrow 1$.
$\lambda_A \leftarrow \{\emptyset\}_{v \in V}$.
while *for some $r \in [p]$, the terminals U_r are not connected within λ_A-covered edges of H* **do**
  Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
  $\lambda_A^j(v) \leftarrow \{i \in \lambda(v) \colon x_{i,v} \geq t_i\}$, for every $v \in V$.
  $\lambda_A \leftarrow \lambda_A^j \cup \lambda_A$.
  $j \leftarrow j + 1$.
return λ_A

Theorem 4.1. *There exists an $O(\log^2 m)$-approximation algorithm for both the min-sum and the min-max connection assignment problems which succeeds with high probability.*

Proof. We show that the above algorithm returns a solution of cost at most $O(\log^2 m) \cdot \texttt{OPT}$ with high probability. We have the following lemmas:

Lemma 4.2. *With high probability, for every $r \in [p]$, the subgraph H connects all terminals in U_r.*

Proof. Fix any $r \in [p]$ and any $S \subset V$ with $S \cap U_r \neq \emptyset$ and $(V \setminus S) \cap U_r \neq \emptyset$. Let $\ell \in \mathbb{N}$, such

that $\ell \le y(\delta(S)) < \ell + 1$. We have:

$$\begin{aligned}
\Pr[|\delta_H(S)| = 0] &= \prod_{e \in \delta(S)} \Pr[Y_e = 0] = \prod_{e \in \delta(S)} (1 - p_e) \\
&\le \prod_{e \in \delta(S)} \exp\{-p_e\} \le \exp\left\{ - \sum_{e \in \delta(S)} \min\{1,\, 4 \ln m \cdot y_e\} \right\} \\
&\le \exp\{-4\ell \ln m\} = \frac{1}{m^{4\ell}}
\end{aligned}$$

where the last inequality uses the LP constraint $y(\delta(S)) \ge 1$ for every U_r-separating cut, so $\sum_{e \in \delta(S)} y_e \ge \ell$.

A cut in G is called an α-minimum cut if its value is less than α times the value of a minimum cut in G. We now make a use of the following Karger's lemma [12, 11]:

Lemma 4.3 (Karger's lemma)**.** *The number of α-minimum cuts in a graph is at most $n^{2\alpha}$.*

Observe that by construction of the LP, every U_r-separating fractional cut has value at least 1, so there are at most $n^{2\ell}$ cuts that are U_r-separating and have fractional value less than ℓ. Applying the union bound over all groups $r \in [p]$ and all U_r-separating cuts:

$$\begin{aligned}
\Pr\big[\exists\, r \in [p],\, \exists\, U_r\text{-sep. cut } S\colon |\delta_H(S)| = 0\big] &\le \sum_{r=1}^{p} \sum_{\ell=1}^{\infty} \sum_{\substack{S \subset V,\, S \cap U_r \neq \emptyset \\ (V \setminus S) \cap U_r \neq \emptyset,\, y(\delta(S)) < \ell+1}} \Pr[|\delta_H(S)| = 0] \\
&\le p \cdot \sum_{\ell=1}^{\infty} n^{2\ell} \cdot \frac{1}{m^{4\ell}} = p \cdot \sum_{\ell=1}^{\infty} \frac{1}{m^{2\ell}} = \frac{p}{m^2 - 1} \\
&= \frac{p}{(m-1)\,(m+1)} \le \frac{1}{m-1},
\end{aligned}$$

where the last inequality uses $p \le n \le m + 1$, since G is connected. Thus, with high probability, no U_r-separating cut is empty in H for any $r \in [p]$. □

Let j be any operation of the while loop. Let H_j be the subgraph of H that is not covered by λ_A at the beginning of the j-th iteration of the while loop. We wish to show, that in expectation, the solution λ_A produced in the j-th iteration of the while loop, call it λ_A^j, covers a significant fraction of the edges in H_j. We have the following lemma:

Lemma 4.4. *We have* $\mathbb{E}\left[\left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|\right] \ge \frac{1-1/e}{3 \ln m} \cdot \mathbb{E}\left[|E(H_j)|\right]$.

Proof. Consider any edge $uv \in E(G)$. Observe that $\mathbb{E}[Y_{uv}] \le 4 \ln m \cdot y_{uv}$. By the definition of the LP we get that:

$$\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \ge y_{uv} \ge \frac{\mathbb{E}[Y_{uv}]}{4 \ln m}.$$

Therefore, the probability that uv is uncovered by λ_A is at most:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-\min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}
\end{aligned}$$

Thus we get $\Pr[\lambda_A(u) \cap \lambda_A(v) \neq \emptyset] \geq 1 - \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}$. We now sum over all edges in H_j to get:

$$\begin{aligned}
\mathbb{E}\left[\left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|\right] &= \sum_{uv \in E(H_j)} \Pr[\lambda_A(u) \cap \lambda_A(v) \neq \emptyset] \\
&\geq \sum_{uv \in E(H_j)} \left\{1 - \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}\right\} \\
&\geq (1 - 1/e) \cdot \sum_{uv \in E(H_j)} \frac{\mathbb{E}[Y_{uv}]}{4 \ln m} \geq \frac{1 - 1/e}{4 \ln m} \cdot \mathbb{E}\left[|E(H_j)|\right]
\end{aligned}$$

where the last inequality is by the linearity of expectation. □

Now, we want to show that the number of iterations of the while loop is at most $O\big(\log^2 m\big)$ with high probability. Let I be the random variable denoting the number of iterations of the while loop. We first bound the expected number of iterations.

Lemma 4.5. $\mathbb{E}[I] \leq \frac{9 \ln^2 m}{1 - 1/e}$.

Proof. Let $\alpha = \frac{1 - 1/e}{4 \ln m}$. By definition, $|E(H_{j+1})| = |E(H_j)| - \left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|$. By linearity of expectation and the previous lemma, $\mathbb{E}\left[|E(H_{j+1})|\right] \leq (1 - \alpha) \cdot \mathbb{E}\left[|E(H_j)|\right]$. Unfolding the recurrence we get that, $\mathbb{E}\left[|E(H_j)|\right] \leq (1 - \alpha)^j \cdot m \leq m e^{-\alpha j}$. By Markov's inequality, $\Pr\{|E(H_j)| \geq 1\} \leq m e^{-\alpha j}$. Let $j^* = \left\lfloor \frac{9 \ln^2 m}{1 - 1/e} \right\rfloor$. We have:

$$\begin{aligned}
\mathbb{E}[I] = \sum_{j=1}^{\infty} \Pr[I \geq j] &= \sum_{j=1}^{j^*} \Pr[I \geq j] + \sum_{j=j^*+1}^{\infty} \Pr[I \geq j] \\
&\leq j^* + \sum_{j=j^*+1}^{\infty} m e^{-\alpha(j-1)} = j^* + m e^{-\alpha j^*} \cdot \frac{e^{\alpha}}{1 - e^{-\alpha}}.
\end{aligned}$$

Since $j^* \geq \frac{2 \ln m}{\alpha} = \frac{8 \ln^2 m}{1 - 1/e}$, we have $m e^{-\alpha j^*} \leq m e^{-2 \ln m} = \frac{1}{m}$, and $\frac{e^{\alpha}}{1 - e^{-\alpha}} \leq \frac{2}{\alpha} = \frac{8 \ln m}{1 - 1/e}$ for $m \geq 2$. Therefore, $\mathbb{E}[I] \leq j^* + \frac{1}{m} \cdot \frac{8 \ln m}{1 - 1/e} \leq \frac{9 \ln^2 m}{1 - 1/e}$ for high enough m. □

Lemma 4.6. $\Pr\left[I \geq \frac{9 \ln^2 m}{1 - 1/e}\right] \leq \frac{1}{m}$.

Proof. Let $\alpha = \frac{1-1/e}{4\ln m}$ and $j^* = \frac{9\ln^2 m}{1-1/e}$. As shown in the proof of Lemma 4.5, $\Pr\{|E(H_j)| \geq 1\} \leq me^{-\alpha j}$. Setting $j = j^* \geq \frac{2\ln m}{\alpha} = \frac{8\ln^2 m}{1-1/e}$ gives $me^{-\alpha j^*} \leq me^{-2\ln m} = \frac{1}{m}$. Hence, $\Pr[I \geq j^*] = \Pr\{|E(H_{j^*})| \geq 1\} \leq \frac{1}{m}$. □

We get that the procedure executes at most $O\big(\log^2 m\big)$ iterations with high probability, so the running time of the algorithm is polynomial with high probability. It remains to show, that with high probability, the cost of the solution returned by the algorithm is at most $O\big(\log^2 m\big) \cdot \texttt{OPT}$. We have:

Lemma 4.7. *With high probability, we have that* $\texttt{COST}(\lambda_A) \leq \frac{27\ln^2 m}{1-1/e} \cdot \texttt{OPT}$.

Proof. Fix an expensive vertex $v \in V$; cheap vertices have cost at most $1 \leq 2\texttt{OPT}$ regardless of I. Let $\mu_v = \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$.

By $X^j_{v,i}$ denote the indicator that $x_{i,v} \geq t_i$ in iteration j. Since different iterations draw independent fresh thresholds, the cost variables $\{c\Big(\lambda^j_A(v)\Big)\}_{j\geq 1}$ are i.i.d. with mean μ_v, and I is independent of the sequence $\Big(c\Big(\lambda^j_A(v)\Big)\Big)_{j\geq 1}$. Applying Wald's identity:

$$\mathbb{E}\left[c(\lambda_A(v))\right] = \mathbb{E}[I] \cdot \mu_v \leq \frac{9\ln^2 m}{1-1/e} \cdot \mu_v.$$

By the weighted Chernoff bound with $\delta = 2$:

$$\Pr\left[c(\lambda_A(v)) \geq \frac{27\ln^2 m}{1-1/e} \cdot \mu_v\right] \leq \left(\frac{e^2}{27}\right)^{\frac{9\ln^2 m}{1-1/e}\cdot\mu_v} \leq e^{-\frac{9\ln^2 m}{1-1/e}\cdot\mu_v} \leq e^{-\frac{9\ln^2 m}{1-1/e}} \leq \frac{1}{m^9},$$

where the last inequality uses $\frac{9\ln^2 m}{1-1/e} \geq 9\ln m$ for $m \geq 2$. By the union bound over all $n \leq m$ expensive vertices, with probability at least $1 - \frac{1}{m^8}$, for every $v \in V$:

$$c(\lambda_A(v)) \leq \frac{27\ln^2 m}{1-1/e} \cdot \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \leq \frac{27\ln^2 m}{1-1/e} \cdot \texttt{OPT}.$$

□

We conclude, that with high probability the algorithm returns a solution which covers all edges in H in at most $O\big(\log^2 m\big)$ iterations and has cost at most $O\big(\log^2 m\big) \cdot \texttt{OPT}$. Since H connects every group U_r internally, the solution returned by the algorithm is a valid solution to the connectivity problem with cost at most $O\big(\log^2 m\big) \cdot \texttt{OPT}$ with high probability. □

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A Weighted Chernoff Bound Derivation

Theorem A.1 ((weighted) Chernoff bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables taking values in $[0,1]$, let $w_1, w_2, \ldots, w_n$ be non-negative weights, such that for every $i \in [n]$ we have that $w_i \in [0,1]$. Let $X = \sum_{i=1}^n w_i \cdot X_i$ and let $\mu = \mathbb{E}[X]$. Then for every $\delta_1 > 0$ and $\delta_2 \in (0,1)$ we have that:*

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu} \qquad \Pr(X \leq (1-\delta_2)\cdot\mu) \leq \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

Proof. We first prove the upper-tail bound for $\delta_1 > 0$. For any $\lambda > 0$, Markov's inequality gives:

$$\Pr(X \geq (1+\delta_1)\cdot\mu) = \Pr\Big(e^{\lambda X} \geq e^{\lambda\cdot(1+\delta_1)\cdot\mu}\Big) \leq \frac{\mathbb{E}\big[e^{\lambda X}\big]}{e^{\lambda\cdot(1+\delta_1)\cdot\mu}}.$$

By independence, we have $\mathbb{E}\big[e^{\lambda\cdot X}\big] = \prod_{i=1}^n \mathbb{E}\big[e^{\lambda\cdot w_i\cdot X_i}\big]$. Fix $i \in [n]$. Since $w_i \cdot X_i \in [0,1]$, by convexity of $y \mapsto e^{\lambda\cdot y}$ on $[0,1]$, for every $y \in [0,1]$, $e^{\lambda\cdot y} \leq (1-y)\cdot e^0 + y\cdot e^{\lambda} = 1 + \big(e^{\lambda}-1\big)\cdot y$. Substituting $y = w_i \cdot X_i$, taking expectation, and using $1 + t \leq e^t$, we obtain:

$$\mathbb{E}\Big[e^{\lambda\cdot w_i\cdot X_i}\Big] \leq 1 + \Big(e^{\lambda}-1\Big)\cdot w_i\cdot\mathbb{E}[X_i] \leq \exp\Big(\Big(e^{\lambda}-1\Big)\cdot w_i\cdot\mathbb{E}[X_i]\Big).$$

Hence:

$$\mathbb{E}\Big[e^{\lambda X}\Big] \leq \exp\left(\Big(e^{\lambda}-1\Big)\cdot\sum_{i=1}^n w_i\cdot\mathbb{E}[X_i]\right) = \exp\Big(\Big(e^{\lambda}-1\Big)\cdot\mu\Big).$$

Therefore, $\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \exp\big(\mu\cdot\big(e^{\lambda}-1-\lambda\cdot(1+\delta_1)\big)\big)$. The function $f(\lambda) = e^{\lambda}-1-\lambda\cdot(1+\delta_1)$ is minimized at $\lambda = \ln(1+\delta_1)$, and so:

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \exp(-\mu\cdot((1+\delta_1)\cdot\ln(1+\delta_1)-\delta_1)) = \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu}.$$

Now let $\delta_2 \in (0,1)$. For any $t > 0$, again by Markov's inequality:

$$\Pr(X \leq (1-\delta_2)\cdot\mu) = \Pr\Big(e^{-tX} \geq e^{-t\cdot(1-\delta_2)\cdot\mu}\Big) \leq \frac{\mathbb{E}\big[e^{-t\cdot X}\big]}{e^{-t\cdot(1-\delta_2)\cdot\mu}}.$$

Applying the same mgf estimate with $\lambda = -t < 0$ yields $\mathbb{E}\big[e^{-t\cdot X}\big] \leq \exp\big(\big(e^{-t}-1\big)\cdot\mu\big)$, so:

$$\Pr(X \leq (1-\delta_2)\cdot\mu) \leq \exp\big(\mu\cdot\big(e^{-t}-1+t\cdot(1-\delta_2)\big)\big).$$

The right-hand side is minimized at $t = -\ln(1-\delta_2)$, which gives:

$$\Pr(X \leq (1-\delta_2)\cdot\mu) \leq \exp(-\mu\cdot(\delta_2+(1-\delta_2)\cdot\ln(1-\delta_2))) = \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

□